Zheyu Qi

PhD Candidate in Finance, University of South Carolina
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Education

University of South Carolina — Columbia, SC
Ph.D. in Business Administration - Finance — *August 2022 - May 2027 (expected)*

Clark University — Worcester, MA
M.S. in Finance, with Distinction — *August 2019 - May 2021*

Inner Mongolia University of Finance and Economics — Hohhot, China
B.S. in Economics in Financial Engineering — *August 2015 - May 2019*

Research Interests

Banking and Financial Institutions, Artificial Intelligence, Neutral language Processing (NLP), Household Finance, International Finance

Working Papers

* *presented by a coauthor; Scheduled presentations are included.*

The Algorithmic Banker: AI Adoption and Consumer Service in Retail Banking

(Job Market Paper, Sole-Authored)

Abstract: We provide the first large-scale evidence on how artificial intelligence (AI) adoption in retail banking affects consumer outcomes. Using CFPB complaint data (2010–2023), we construct a bank-county-quarter panel linking the universe of consumer complaints to bank-level AI patent filings and show that AI adoption reduces aggregate complaint rates by approximately 30 percent relative to the sample mean. An event-study design confirms parallel pre-trends and reveals that complaint reductions emerge with a two-year lag consistent with gradual technology deployment; propensity score matching and a Bartik-style instrumental variable corroborate a causal interpretation. However, these gains conceal opposing effects across bank services. AI eliminates operational errors in transaction processing and credit reporting while generating new grievances through opaque algorithmic decisions and rigid automation in credit and customer service. AI adoption also improves resolution quality. Adopting banks deliver a higher share of substantive, non-monetary remediation and fewer outright dismissals, indicating that the complaint decline reflects genuine service improvement rather than complaint suppression. Finally, applying dictionary-based textual analysis to complaint narratives, we show that AI adoption reduces discrimination-related complaints.

***Presentations*: 2026 FMA, 3rd AI in Finance, 2026 SFA, University of South Carolina**

From Soft to Hard: How AI Transforms Bank Information Production in Mortgages

(with Allen N. Berger, Omrane Guedhami, Raluca A. Roman)

Abstract: AI is transforming how banks produce information for credit decisions, but existing research treats the process as a black box. We identify the mechanism: AI converts soft information — difficult-to-verify, judgment-dependent borrower signals — into standardized hard-information signals. Linking 1.8 million HMDA mortgage applications (2018–2024) to bank-level AI patent and employee measures, we show that AI adoption reduces denials by 4.7 percentage points. The reduction is concentrated entirely in soft denials, which capture processing-related frictions rather than clear credit-quality failures; hard-information denials are largely unchanged. Among denied applicants, soft-information denials fall by 12.4 percentage points, a 34.5% decline. Effects are strongest where soft information is most salient: jumbo loans, informationally advantaged markets, and proprietary underwriting systems. Non-bank mortgage companies deploy AI talent at higher intensity than

banks yet exhibit no parallel effects, as they lack the relationship-based infrastructure that generates the soft information AI requires as input. AI also reduces loan pricing at banks, with no corresponding effects at non-bank lenders. Overall, the evidence indicates that AI converts difficult-to-process borrower information into standardized underwriting signals.

Banking on Artificial Intelligence: Information Hold-Up and the Transformation of Relationship Lending

(with Allen N. Berger, Omrane Guedhami, Raluca A. Roman)

Abstract: We find bank AI adoption raises corporate loan spreads by 9–13 bps, with propensity-score matching and Bartik-style instruments confirming robustness. AI converts soft information into proprietary hard information, deepening rather than reducing information asymmetry. Relationship price discounts erode at AI-adopting banks, yet relationships persist through non-price terms – smaller, shorter loans with fewer covenants – shifting function from spread reduction to rollover discipline. Machine learning (1995–2010) reprices public borrowers; deep learning and NLP (2011–2023) shift burdens toward private borrowers. Borrowers adopting AI face lower spreads, but this discount vanishes at AI-adopting banks, consistent with rent extraction over efficiency gains.

***Presentations*: 2026FMA, EUROFIDAI-ESSEC Paris, EasternFA 2026, 3rd Ai in Finance, University of South Carolina**

The Paycheck Protection Program (PPP) from the Small Business Perspective: Did the PPP Help Alleviate Financial and Economic Constraints?

(with John Ampong, Allen N. Berger, Paul G. Freed, Zheyu Qi, and Jonathan A. Scott)

-- R&R in JBF

Abstract: We employ two novel datasets to evaluate whether the Paycheck Protection Program (PPP) met short-term and longer-term goals. One dataset matches PPP with small business members of the National Federation of Business (NFIB) that report their financial constraints and their employees' economic constraints. We find more alleviation of both constraints for PPP recipients, consistent with short-term goals. We employ data on post- versus pre-crisis county employment, wages, and new businesses to test the longer-term goal of improved community crisis recoveries from the short-term constraint relief. We find to the contrary, worse longer-term recoveries in counties with more short-term constraint relief.

Presentations*: 2022 SFA*, 2022 FMA*, Temple University*, University of South Carolina

Lending in the Age of Cyber Threats: How do lenders Price Loans?

(Solo-Authored)

-AFS 2025 Conference - Outstanding Doctoral Research Paper Award

Abstract: We examine how lenders price the emerging risks of data breaches in the syndicated loan market. We find that bank lenders increase loan spreads for firms that experienced a data breach during the loan underwriting process. This increase is particularly pronounced for credit lines, rather than term loans. These findings support the Bank Specialness Hypothesis, suggesting that banks maintain a significant informational advantage in pricing new risk factors compared to non-bank lenders. Our results also indicate that firms experiencing a data breach can substantially mitigate these pricing effects through two channels: strong borrower-lender relationships and proactive cybersecurity risk management. This finding complements existing relationship lending studies by incorporating cybersecurity risk.

***Presentations*: 51st EBES Conference Annual Meeting 2025, AFS Annual Conference 2025, University of South Carolina**

Work in Progress

Artificial Intelligence and the Geography of Credit: Evidence from Global Syndicated Lending (with Omrane Guedhami, Raluca Roman)

Status: We study how does AI adoption by global banks affect syndicated lending across borders, and does this effect vary systematically with institutional distance and regulatory differences between lender and borrower countries

Boomerang Credit: Do Anti-ESG Laws Reallocate Bank Corporate Credit to Fossil-Fuel Firms? (with Allen Berger, Paola Dorazio, Raluca Roman)

Status: We study whether state-level anti-ESG laws, which restrict public entities from doing business with financial institutions perceived as boycotting fossil-fuel or firearms companies, spill over from the municipal finance market into the corporate credit market.

Teaching Experience

Instructor FINA 365 Corporate Finance (Undergraduate), 2025 Fall (Evaluations:**4.0/5.0**)

Selected Students Comments:

-*Professor really cares and takes time to talk out theories, not only showing on slide shows. I think she believes that if you understand it you can talk about it and that leads us to be able to understand more since she makes us talk about what we learn very effortlessly.*

-*She actually cares about her students and wants them to learn. She makes sure everyone is on pace during class and slows down and helps students if needed.*

-*She is very down-to-earth and knows how to create a good learning environment.*

-*The lecture structure resonating with me well. Being a smaller class, we were encouraged to participate more in class which helps me hold the information longer.*

Teaching Assistant

2025,2023 Spring FINA 469 Investment Analysis and Portfolio Management (Undergraduate)
2024, Spring FINA 365 Corporate Financial Analysis (Undergraduate)

Selected Research Assistance

For Allen Berger and Sergey Tsyplakov, University of South Carolina	2024 - 2025
For Da Ke and Sergey Tsyplakov, University of South Carolina	2023 - 2024
For Allen Berger, DongMei Li, University of South Carolina	2022 - 2023

Academic Services

Conference Session Chair

SFA (x2)	2026
FMA	2026
EasternFA	2026
FMA	2025

Conference Discussions

SFA discussant	2026
FMA discussant	2026
EasternFA discussant	2026

FMA discussant (x2) 2025

Internal Services

Exam Coordinator, Department of Finance 2024 - 2025

Awards and Honors

University of South Carolina Graduate School Travel Grant	2026
Outstanding Doctoral Research Paper Award. AFS Conference	2025
Special Scholarship- Darla Moore Doctoral Fellows Award	2025
First Year Paper Reward	2023
Doctoral Fellowship. Darla Moore School of Business	2022 - 2027
Clark University- School of Management Outstanding Student Award (Top 1%)	2021
China National Scholarship (Top 1%)	2019

Skills

Technical Skills: LLM (Large Langrage Model). NLP (Natural Langrage Processing), STATA, Python, SAS, R, LaTeX, Bloomberg Terminal

Languages: Chinese (native), English (fluent)

Industry Experiences

JW Real Estate Services Company - Marketing Department	Boston, United States Aug 2021 – Jun 2022
Guangxia Investment Management Co., Ltd. - Quantitative Trading Department	Shanghai, China Jan 2021 – Mar 2021
China Merchant Securities - IPO Department	Shanghai, China Aug 2020 – Dec 2020
China Merchant Securities - ABS Department	Beijing, China Jun 2020 – Jul 2020
ADS Securities Hong Kong Ltd - Foreign Exchange Transactions Department	Hong Kong, China Jul 2017 – Sep 2017

References

Allen N. Berger (Co- Chair)
Carolina Distinguished Professor – Montague Osteen Jr. Professor of Banking and Finance,
Darla Moore School of Business, University of South Carolina
E-mail: aberger@moore.sc.edu

Eric A. Powers (Co- Chair)
Chair, Department of Finance
Darla Moore School of Business, University of South Carolina
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Federal Reserve Bank of Philadelphia
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